Predicting HDB Monthly Rent in Singapore

CS5228 Knowledge Discovery and Data Mining

Progress Report · 26 September 2026

Team 29

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Research question

What monthly rent should we expect for an HDB flat at a given date?

Inputs

Flat type, floor area, town,
block location and approval month

Output

Predicted monthly rent
in Singapore dollars

Our first aim is a transparent, time-aware baseline. We will then test whether auxiliary location and economic data improve it.

Data inventory

150,000

labeled training records

Jan 2021 to Mar 2025

50,000

unlabeled test records

Mar 2025 to Jul 2026

Core fields

Date, address, flat type and model, area, lease year, rent

Auxiliary files

HDB blocks, MRT stations, schools, malls, COE prices and stocks

Target: MONTHLY_RENT (SGD). The test file has no target column.

Data quality decisions

01 **No missing cells in the core tables**

All 150,000 training rows and 50,000 test rows have values in their supplied columns.

02 **Two fields carry no variation**

FURNISHED is always “yes” and FEE is always zero; neither is useful in the first model.

03 **Normalize flat type labels**

Training data mixes “4 room” and “4-room”; test data uses the hyphenated form.

04 **Keep repeated rows for now**

425 exact duplicate training rows may still represent different rental approvals.

Forward validation design

A random holdout would mix early and late rental markets. We use the latest six labeled months as a forecast check.

Model fitting

132,802 rows

Jan 2021–Sep 2024

Validation

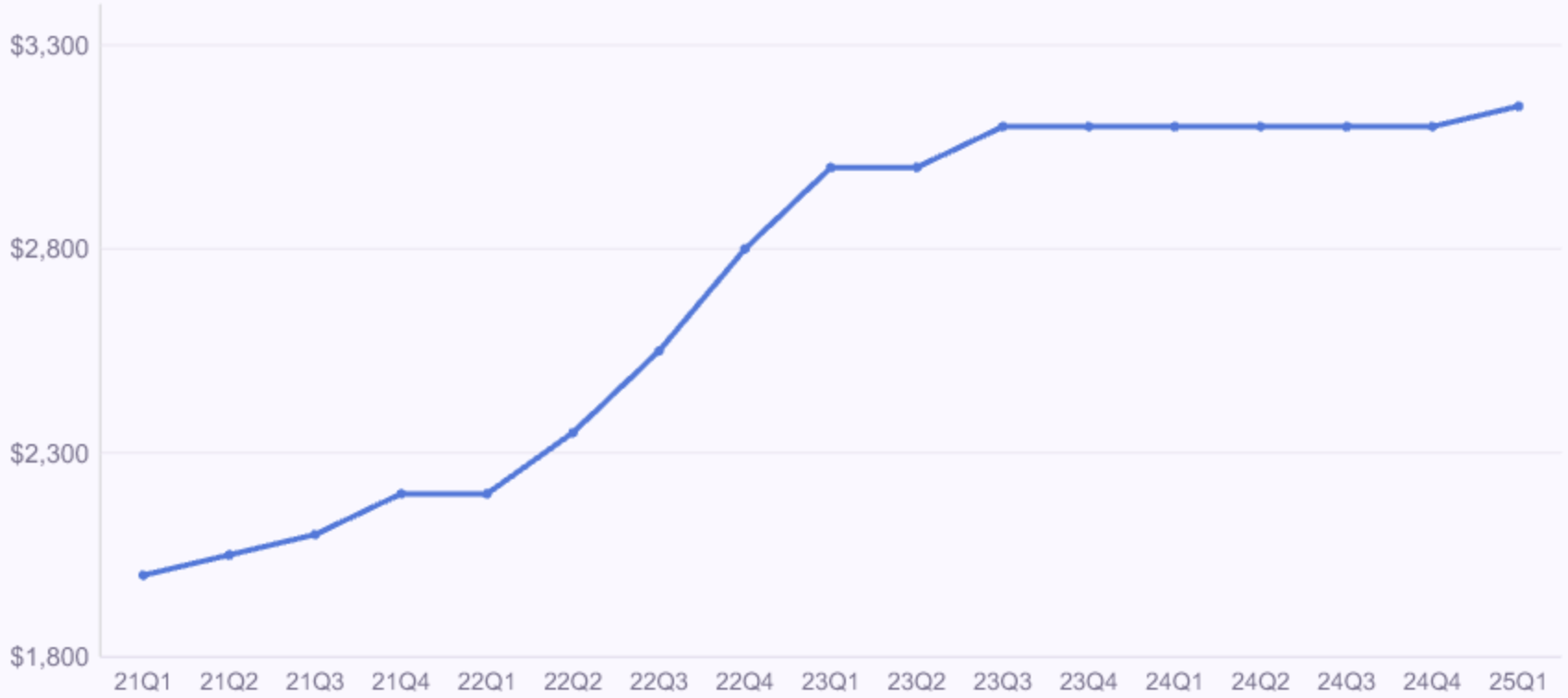
17,198 rows

Oct 2024–Mar 2025



Kaggle scores predictions by RMSE. We also report MAE to describe the typical absolute error in SGD.

Rents rose sharply through 2023



SGD 2,000

2021 Q1 median

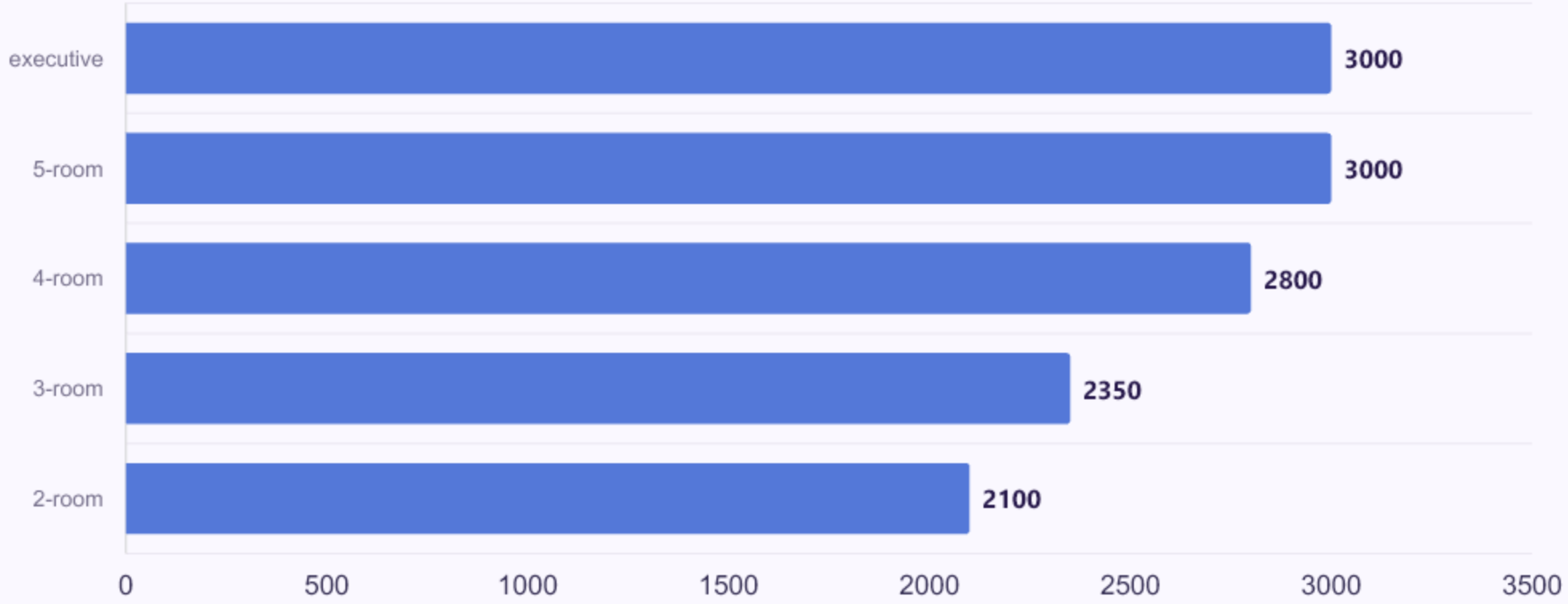
SGD 3,150

2025 Q1 median

+57.5% over this period

Quarterly median approved rent, SGD per month. Values are not adjusted for changes in flat mix.

Rent differs across flat types



The 1-room category has only 42 training records and is omitted from this chart.

Block data adds spatial context

100%

match rate for both train and test addresses

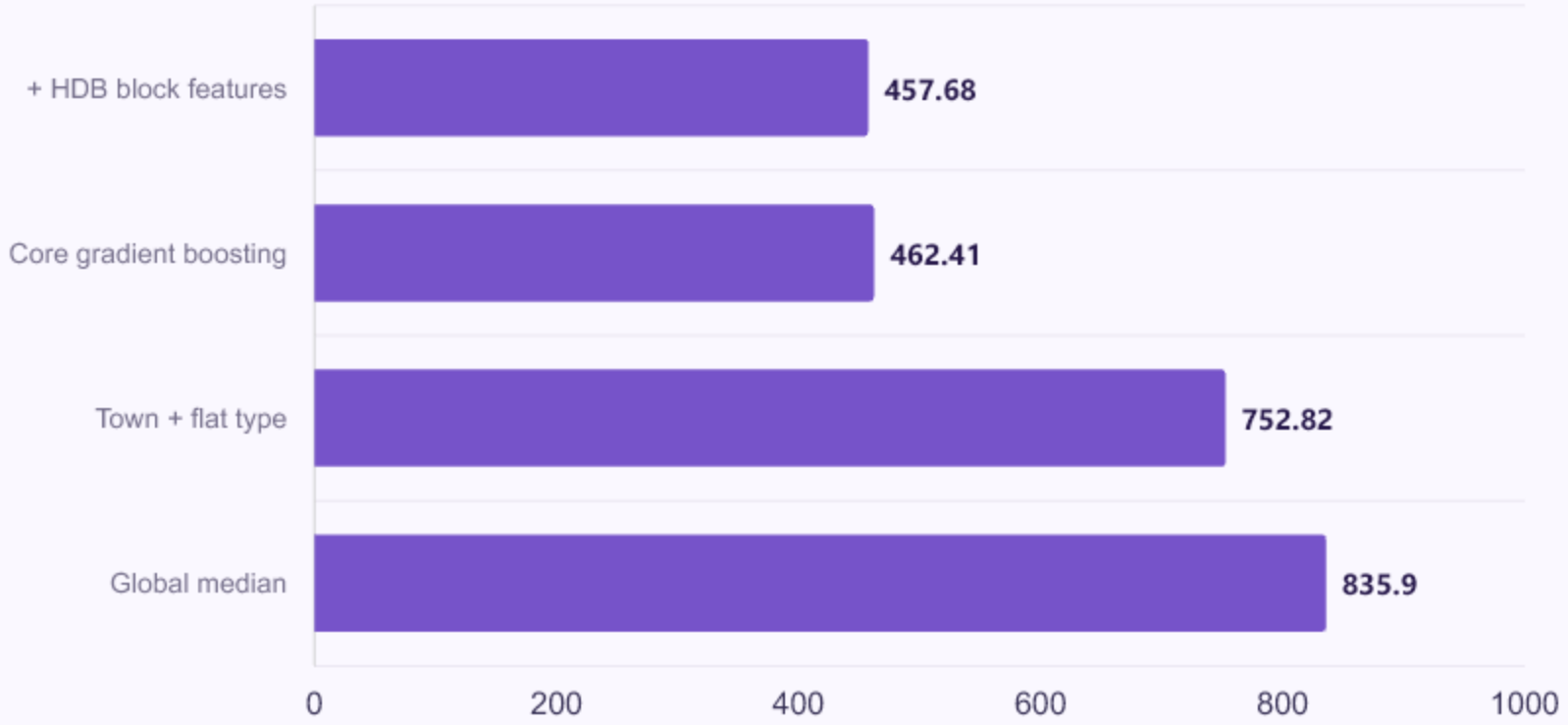
Added features

Latitude and longitude
Maximum floor
Building age
Region

RMSE changes from SGD 462.41 to 457.68 when these features enter the first gradient boosting model.

A 1.0% improvement on this holdout; we will test whether it persists across other periods.

Preliminary validation results



Best model

SGD 457.68

RMSE

SGD 334.55

MAE

The median baselines lag the later, higher-rent market.

Validation: 17,198 records from Oct 2024 through Mar 2025. Lower is better.

Current error patterns and limits



Executive flats

SGD 394 MAE

852 validation rows

This group is harder than the larger 3-, 4- and 5-room groups. We need to inspect high-error cases before choosing new features.

Errors by approval month stay within SGD 326–345 MAE in this six-month window.

Next steps: strengthen evaluation

01 **Match the official RMSE metric**

Keep the local evaluation and Kaggle submission format aligned: Id and Predicted.

02 **Diagnose the historical rent shift**

Across four six-month windows, model RMSE fell from SGD 622 to 458. Test why older windows were harder.

03 **Audit large errors**

Review misses by town, flat type, approval month and rent band; inspect suspicious records.

04 **Measure longer horizons**

The test period runs into Jul 2026, much farther than the current six-month holdout.

Next steps: complete the project

Feature work

Measure distance to existing MRT stations, schools and malls. Join COE and stock indicators by approval month without using future values.

Model comparison

Compare stronger regression methods with the same forward splits. Keep a simple baseline and record each feature group’s contribution.

Final deliverables

Prepare the reproducible code, Kaggle submission and an eight-page scientific report with team contributions.